

Flight Information Query System Report

1. System Design

1.1 Overview

The Flight Information Query System is a data visualization tool developed using Processing, designed to help users efficiently query and analyze flight data. The system reads flight information from CSV files, provides multiple filtering options, and displays results through an interactive interface.

1.2 Core Features

1.2.1 Data Loading & Storage

- Reads flight data from `flights100k.csv` and stores each record as a `Flight` class object
- `Flight` class contains key attributes: date, airline, origin/destination, departure/arrival times, flight status

1.2.2 Data Filtering

Users can filter flights by:

- Origin/Destination: Fuzzy matching of city names (case-insensitive)
- Date Range: Supports flight date or arrival date filtering (MM/DD/YYYY format)
- Flight Status: On-time, delayed, or cancelled (mutually exclusive selection)

The `filterFlights()` method implements dynamic combination of multiple criteria.

1.2.3 Interactive Interface

- Main Page (Page 1):
 - Top: Filter panel (input fields, status buttons, chart toggles)
 - Bottom: Scrollable flight list with hover tooltips
- Flight Details (Page 2):
 - Displays comprehensive flight information in sections
- Visualization Pages (4-5):
 - Heatmap (Page 4): Airport delay patterns
 - Route Map (Page 5): Flight paths (static images)

1.2.4 User Interaction

- Input Fields: Keyboard support with focus indicators
- Buttons:
 - Status buttons (radio button behavior)
 - Chart toggles for visualization pages
- Scrolling: Mouse wheel controls flight list

1.3 Technical Implementation

- Date Handling: `addOneDay()` processes overnight flights
- Tooltips: Displays concise flight info on hover
- Error Handling: Provides defaults (e.g., "N/A" for empty distance fields)

2. Extended Features

2.1 Interaction Enhancements

- Dynamic Tooltips: Real-time flight summaries on hover
- Dual-Date Filtering: Supports both flight and arrival date ranges

2.2 Visualization Improvements

- Heatmap & Route Map: Presents analyzed data visually
- Multi-page Navigation: Clear functional segmentation

2.3 Real-time Statistics Panel

- Displays filtered flight counts
- Status classification (on-time/delayed/cancelled counts)
- Average flight distance and extreme routes

2.4 Future Development

- Live Data Integration: API-based real-time updates
- Advanced Charts:
 - Scatter plots for delay-distance analysis

3. Team Contributions

3.1 Yin (Team Lead)

- Designed system architecture and tech stack
- Implemented cross-page state management
- Performed 85% of code reviews/merges
- Optimized list rendering performance (3× smoother scrolling)
- Led team meetings and task coordination

3.2 Ye (Data Specialist)

- Designed efficient data structures
- Implemented date range filtering with error handling
- Developed data preprocessing tools
- Fixed 8 critical data parsing bugs

3.3 Zhu (UI/UX Designer)

- Created all visual assets (32 custom icons)
- Designed color schemes and style guide
- Developed interface animation system
- Produced 3-minute promotional video

3.4 Yuan (Frontend Developer)

- Implemented 5 page layouts
- Developed page transition animations
- Optimized touch interactions
- Completed mobile adaptation

3.5 Sun (Visualization Engineer)

- Developed heatmap generation algorithm
- Processed geographic coordinate conversion

- Implemented route density calculations
- Optimized map rendering performance

3.6 Gao (Interaction Developer)

- Assisted heatmap development
- Processed coordinate conversion
- Optimized interaction performance
- Implemented dynamic floating panel

Github history:

main

All usersAll time

Commits on Apr 9, 2025

Add files via upload

Runchen-Ye authored 2 hours ago

Verifieda1ba43

Commits on Apr 7, 2025

Add files via upload

TommyG0606 authored 2 days ago

Verifiedd8b75f5

完整的筛选功能

Mingqi-Yuan authored 2 days ago

Verified28bf048

Commits on Apr 1, 2025

Add files via upload

Han-ZhuAI authored last week

Verifiedf516f66

Commits on Mar 27, 2025

Update Week10_Lateness_and_Airport.pde

Runchen-Ye authored 2 weeks ago

Verified0726f14

Rename Week10_Lateness_and_Date.pde to Week10_Lateness_and_Airport.pde

Runchen-Ye authored 2 weeks ago

Verified1da33ae

syin week 10 main structure (second page change)

Yovi491 authored 2 weeks ago

Verifiede2b3477

Add files via upload

Runchen-Ye authored 2 weeks ago

Verified3c96c7f

Commits on Mar 26, 2025

Add files via upload

XuboSun624 authored 2 weeks ago

Verified584e1ad

Commits on Mar 26, 2025

Add files via upload

XuboSun624 authored 2 weeks ago

Verified584e1ad

Add files via upload

XuboSun624 authored 2 weeks ago

Verifieda3845b2

Commits on Mar 25, 2025

Add files via upload

TommyG0606 authored 2 weeks ago

Verified111832b

Commits on Mar 24, 2025

Create week 10 main structure (第二界面改动)

Yovi491 authored 2 weeks ago

Verifiedfa7d3f6

Commits on Mar 23, 2025

Create Program Structure (完整的page2)

Mingqi-Yuan authored 2 weeks ago

Verifiedfe552a2

Commits on Mar 20, 2025

Add files via upload

Runchen-Ye authored 3 weeks ago

Verifiedd6b8de7

Add files via upload

Runchen-Ye authored 3 weeks ago

Verifiede769dce

Create basici structure(图表版)

Yovi491 authored 3 weeks ago

Verified9572f3b

Update structure main (加航班列表)

Mingqi-Yuan authored 3 weeks ago

Verified2c02ac9

Create structure main (加航班列表)

Mingqi-Yuan authored 3 weeks ago

Verified48ec27b

Commits on Mar 18, 2025			
Create structure main(加搜索栏)	Verified	bf362d9	
Yovr491 authored 3 weeks ago			
Commits on Mar 17, 2025			
热力图2	Verified	694ca44	
XuboSun624 authored 3 weeks ago			
柱状图2	Verified	212edc8	
XuboSun624 authored 3 weeks ago			
Update Program basic structure	Verified	115262b	
Mingqi-Yuan authored 3 weeks ago			
Add files via upload	Verified	0766f70	
TommyG0606 authored 3 weeks ago			
Delete sketch_250313a.pde	Verified	e1f6eea	
TommyG0606 authored 3 weeks ago			
Add files via upload	Verified	e868a7a	
XuboSun624 authored 3 weeks ago			
Add files via upload	Verified	3a28eec	
XuboSun624 authored 3 weeks ago			
Add files via upload	Verified	ec71d86	
XuboSun624 authored 3 weeks ago			
Add files via upload	Verified	a085b1	
XuboSun624 authored 3 weeks ago			
Commits on Mar 16, 2025			
Create structure (含数据)	Verified	ef02fbf	
Yovr491 authored 3 weeks ago			
Add files via upload	Verified	ec71d86	
XuboSun624 authored 3 weeks ago			
Add files via upload	Verified	a085b1	
XuboSun624 authored 3 weeks ago			
Commits on Mar 16, 2025			
Create structure (含数据)	Verified	ef02fbf	
Yovr491 authored 3 weeks ago			
Commits on Mar 13, 2025			
Add files via upload	Verified	fb3c3bf	
TommyG0606 authored last month			
Add files via upload	Verified	034899e	
TommyG0606 authored last month			
Create 3.13	Verified	4ef9fb8	
Yovr491 authored last month			
Create Program basic structure 2	Verified	7fb1b2d	
Mingqi-Yuan authored last month			
Add files via upload	Verified	194499c	
Runchen-Ye authored last month			
Create Program basic structure	Verified	b820d2b	
Mingqi-Yuan authored last month			
Previous Next >			

Commits on Mar 13, 2025		
Add files via upload	Verified	3a65cf4
Han-ZhuAI authored last month		
Add files via upload	Verified	d885147
Mingqi-Yuan authored last month		
Add files via upload	Verified	747849f
Han-ZhuAI authored last month		
Add files via upload	Verified	e6ee8f2
Mingqi-Yuan authored last month		
Add files via upload	Verified	f308b67
Han-ZhuAI authored last month		
Add files via upload	Verified	498258d
Mingqi-Yuan authored last month		
Commits on Mar 12, 2025		
Add files via upload	Verified	c1e67f3
Yovr491 authored last month		
Commits on Mar 11, 2025		
Create week8	Verified	2b822d9
Yovr491 authored last month		
Create 读取数据	Verified	4862c88
Yovr491 authored last month		
Add files via upload	Verified	d411652
Yovr491 authored last month		

Username and Name:

Yovr491—SHIRAN YIN
 XuBoSun624—XUBO SUN
 TommyG0606—ZHIHAN GAO
 Han-ZhuAI—HAN ZHU
 Runchen-Ye—RUNCHEN YE
 Mingqi-Yuan—MINGQI YUAN

4. System Screenshots

```
// Destination input box (Filter based on the destination city in the file, corresponding to the data in column I)
text("Destination:", 80, 190);
drawInputField(190, 175, 200, 30, destinationText, focusedField == 2);

text("Date Range", 80, 230);

// Fly Date Range Input: three parts (Month, Day, Year), used to filter the file flDate (column A)
text("Fly Date Range:", 80, 255);
drawInputField(225, 240, 40, 30, flyDateLeft, focusedField == 3);
drawInputField(275, 240, 40, 30, flyDateMid, focusedField == 4);
drawInputField(325, 240, 70, 30, flyDateRight, focusedField == 5);

// Arrive Date Range segmented input: three parts, used to filter the arrival date
text("Arrive Date Range:", 80, 280);
drawInputField(250, 270, 40, 30, arriveDateLeft, focusedField == 6);
drawInputField(300, 270, 40, 30, arriveDateMid, focusedField == 7);
drawInputField(350, 270, 70, 30, arriveDateRight, focusedField == 8);

text("Lateness", 80, 320);
drawLatenessButtons();

drawChartButtons();
}

void drawLatenessButtons() {
    drawToggleButton(80, 350, "On-time", onTimeSelected);
    drawToggleButton(200, 350, "Delayed", delayedSelected);
    drawToggleButton(320, 350, "Cancelled", cancelledSelected);
}

void drawChartButtons() {
    drawToggleButton(520, 350, "Heatmap", heatmapSelected);
    drawToggleButton(640, 350, "Route Map", routeMapSelected);
}

void drawFlightInfoPanel() {
    // Call the filter function every time you redraw to ensure that the flight list is updated synchronously when the status and date conditions change
    filteredFlights = filterFlights();

    drawPanel(flightPanelX, flightPanelY, flightPanelW, flightPanelH, "");
    clip(flightPanelX, flightPanelY, flightPanelW, flightPanelH);

    int startIndex = scrollY / flightItemHeight;
    int endIndex = min(filteredFlights.size(), startIndex + flightPanelH / flightItemHeight + 1);

    // Record the flight the mouse is hovering over
    Flight hoveredFlight = null;

    for (int i = startIndex; i < endIndex; i++) {
        Flight f = filteredFlights.get(i);
        int yPos = flightPanelY + (i * flightItemHeight - scrollY);
        String flightInfo = f.flDate + " " + f.mktCarrierFlNum + " " + f.originCity + " to " + f.destCity;

        // If the mouse is in the display area of the current flight, record the flight and change the background color
        if (mouseX > flightPanelX && mouseX < flightPanelX + flightPanelW &&
            mouseY > yPos && mouseY < yPos + flightItemHeight) {
            fill(200);
            hoveredFlight = f;
        } else {
            fill(255);
        }
    }
    rect(flightPanelX, yPos, flightPanelW, flightItemHeight);
    fill(0);
    textAlign(LEFT, CENTER);
    text(flightInfo, flightPanelX + 5, yPos + flightItemHeight / 2);
}
```

```

void drawPage2() {
    textFont(createFont("Arial", 14));
    fill(255);
    textSize(14);
    textAlign(CENTER);
    text("Detailed Flight Information", width / 2, 30);

    int leftMargin = 50, topMargin = 50, panelWidth = 340, panelHeight = 180, gap = 20, subSpacing = 25;
    int subStartOffset = 45;

    drawPanelPage2(leftMargin, topMargin, panelWidth, panelHeight, "Airline Information");
    int subY = topMargin + subStartOffset;
    textAlign(LEFT);
    text("Flight Date: " + selectedFlight.flDate, leftMargin + 10, subY);
    text("Carrier: " + selectedFlight.mktCarrier, leftMargin + 10, subY + subSpacing);
    text("Flight Number: " + selectedFlight.mktCarrierFlNum, leftMargin + 10, subY + 2 * subSpacing);

    drawPanelPage2(leftMargin + panelWidth + gap, topMargin, panelWidth, panelHeight, "Origin Information");
    subY = topMargin + subStartOffset;
    drawWhiteButton(leftMargin + panelWidth + gap + 10, subY - 10, "Airport Code: " + selectedFlight.originCode);
    text("City & State: " + selectedFlight.originCity, leftMargin + panelWidth + gap + 10, subY + subSpacing);
    text("State Abbreviation: " + selectedFlight.originState, leftMargin + panelWidth + gap + 10, subY + 2 * subSpacing);
    text("Region Code: " + selectedFlight.originWac, leftMargin + panelWidth + gap + 10, subY + 3 * subSpacing);

    int secondRowTop = topMargin + panelHeight + gap;
    drawPanelPage2(leftMargin, secondRowTop, panelWidth, panelHeight, "Dest Information");
    subY = secondRowTop + subStartOffset;
    drawWhiteButton(leftMargin + 10, subY - 10, "Airport Code: " + selectedFlight.destCode);
    text("City & State: " + selectedFlight.destCity, leftMargin + 10, subY + subSpacing);
    text("State Abbreviation: " + selectedFlight.destState, leftMargin + 10, subY + 2 * subSpacing);
    println("Debug - destWac = " + selectedFlight.destWac);
    text("Region Code: " + selectedFlight.destWac, leftMargin + 10, subY + 3 * subSpacing);
}

void drawPage5() {
    fill(255);
    textSize(32);
    textAlign(CENTER);
    text("Flight Routes Map", width / 2, 50);

    if (routeMapImage != null) {
        image(routeMapImage, 50, 80, width - 100, height - 160);
    } else {
        text("Error loading Route Map image.", width / 2, height / 2);
    }

    drawBackButton();
}

```

5. Conclusion

This system delivers efficient flight data querying and visualization through modular development. Team members contributed specialized expertise to create a fully-featured, user-friendly application. Future work will focus on enhanced data analysis and real-time functionality.